TAKSHASHILA UNIVERSITY

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ADVANTAGES OF EXCEPTION HANDLING (IN JAVA):

Prevents Program Crash

- Exception handling stops the program from suddenly crashing when an error occurs. Instead, it handles the error and continues execution.

Maintains Normal Flow of Program

- It allows the program to continue running even after an exception occurs.

Separates Error Handling Code from Normal Code

- The main program logic stays clean because error handling is written in try, catch, and finally blocks.

Improves Program Reliability

- Programs become more stable and reliable because unexpected errors are properly managed.

Helps in Debugging

- Exceptions provide error messages and stack traces that help developers find and fix problems easily

Allows Custom Error Messages

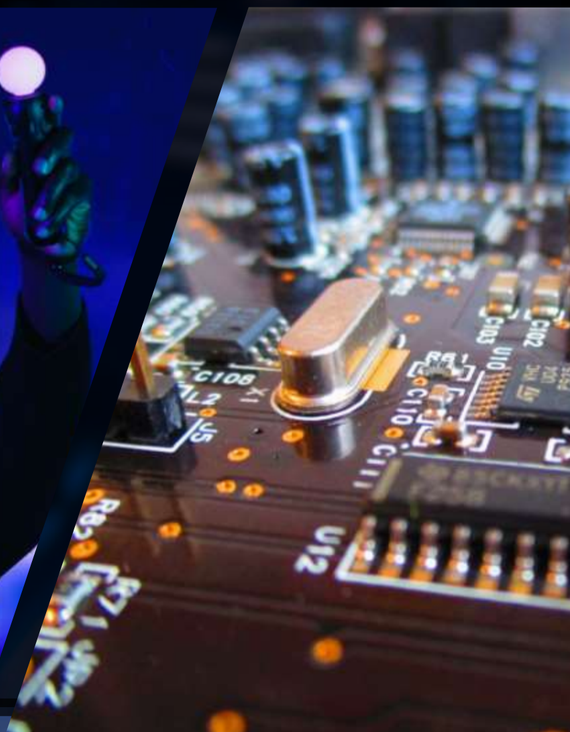
- Developers can create their own exceptions to give meaningful error messages.

Resource Management

- Using finally or try-with-resources ensures resources like files and database connections are properly closed.

Example (Java):

```
try {
   int a = 10;
   int b = 0;
   int c = a / b;
} catch (ArithmeticException e) {
   System.out.println("Cannot divide by
zero");
}
```



Improves Program Readability

- Using try, catch, and finally makes the code easier to understand and organize.

Handles Different Types of Errors

- Multiple catch blocks can handle different exceptions separately.

Graceful Error Handling

- Instead of stopping the program abruptly, it shows user-friendly messages.

Supports Hierarchy of Exceptions

- Java provides a structured exception hierarchy, making it easier to handle related errors.

Propagation of Exceptions

- Exceptions can be passed from one method to another using the throws keyword.

Better Application Security

- It prevents sensitive system details fro being exposed when an error occurs.

Encourages Robust Programming

- Developers think about possible errors and handle them properly.

Allows Logging of Errors

- Exceptions can be logged to files or monitoring systems for later analysis.

Improves User Experience

- Users receive meaningful error messages instead of system crashes.

Simplifies Large Application Management

- In large programs, exception handling helps manage errors in a structured way.